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Abstract

When the development of the J-2 engine for the Saturn V second and third stages was completed, the National Aeronautics and Space Administration (NASA) retained the talents of the contractor personnel for flight anomaly support by sponsoring the design of a simplified J-2 engine system. The J-2S engine was the product of this design and development program. This represented a unique opportunity to redesign an engine immediately upon the completion of producing an operational system. The product was an innovative tap-off cycle engine with improved performance and thrust capability with reduced engine complexity and lower manufacturing costs. This engine has been studied for application in future launch systems requiring 265,000-lb thrust class engines. This paper describes the engine system, its operating features, some basic programmatic considerations for the reinitiation of production of this engine, as well as the uprating potential of this system.

Introduction

Following the completion of development and certification of the J-2 oxygen/hydrogen rocket engine for Saturn vehicles, the NASA initiated efforts to develop an increased thrust, higher performance, yet simpler engine to replace the J-2 engine for Saturn launches following the Apollo program. This effort allowed the J-2 design and development team an opportunity to build on their learning in the J-2 program and produce an engine that addressed all of their previous design concerns in a simpler, more robust engine system known as the J-2 simplified or, as more commonly abbreviated, the J-2S.

Engine Description

The J-2S engine (Fig. 1) is a 265,000-lb vacuum thrust engine (compared to 230,000 lb for the J-2) that delivers 436 sec vacuum thrust (J-2: 427 sec) with a nozzle expansion ratio of 40 (J-2: 27.5) and operates at a chamber pressure of 1,200 psi (J-2: 750 psi). Physically it is an identical replacement of the J-2. The most unique feature of the J-2S is that it incorporates a main chamber tap-off cycle which eliminates the need for a gas generator. Hot gas from the main chamber is diluted with cold hydrogen to achieve the desired turbine inlet temperature. After expansion in the turbines, the hot gas is injected into the nozzle downstream of the throat, the same as the hot gas in the J-2 gas generator cycle engine. A schematic of the J-2S is shown in Fig. 2.

Another simplification for the J-2S was the adoption of a centrifugal fuel turbopump to replace the J-2's axial turbopump. This allowed for greater operating range and greater insensitivity from propellant inlet condition excursions. This change simplified the engine chill procedure considerably.

The J-2's hydrogen spin start bottle was replaced with solid propellant turbine spinners. These solid propellant gas generators (SPGGs) were developed and tested successfully in simulated three start missions. The SPGGs were automatically switched in sequence to provide turbopump spin start upon receipt of the vehicle start signal. This change eliminated all of the thermal conditioning issues associated with the former start bottle system.

Other simplifications to the system were realized by eliminating bellows from the high-pressure cryogenic ducts along with other producibility improvements to most components. The only additional complexity to the J-2S was the addition of the tap-off ports to the combustion chamber. Figure 3 shows a sketch of the tap-off port configuration. These ports were hand brazed for the J-2S engines tested, however in a production environment, a less labor-intensive casting would be used.

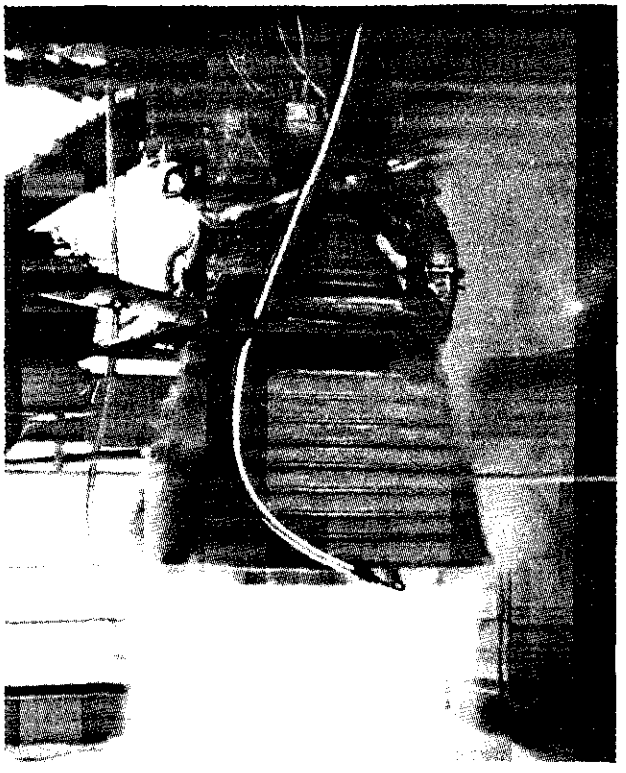


Fig. 1. J-2S in a Sea Level Hot-Fire Test

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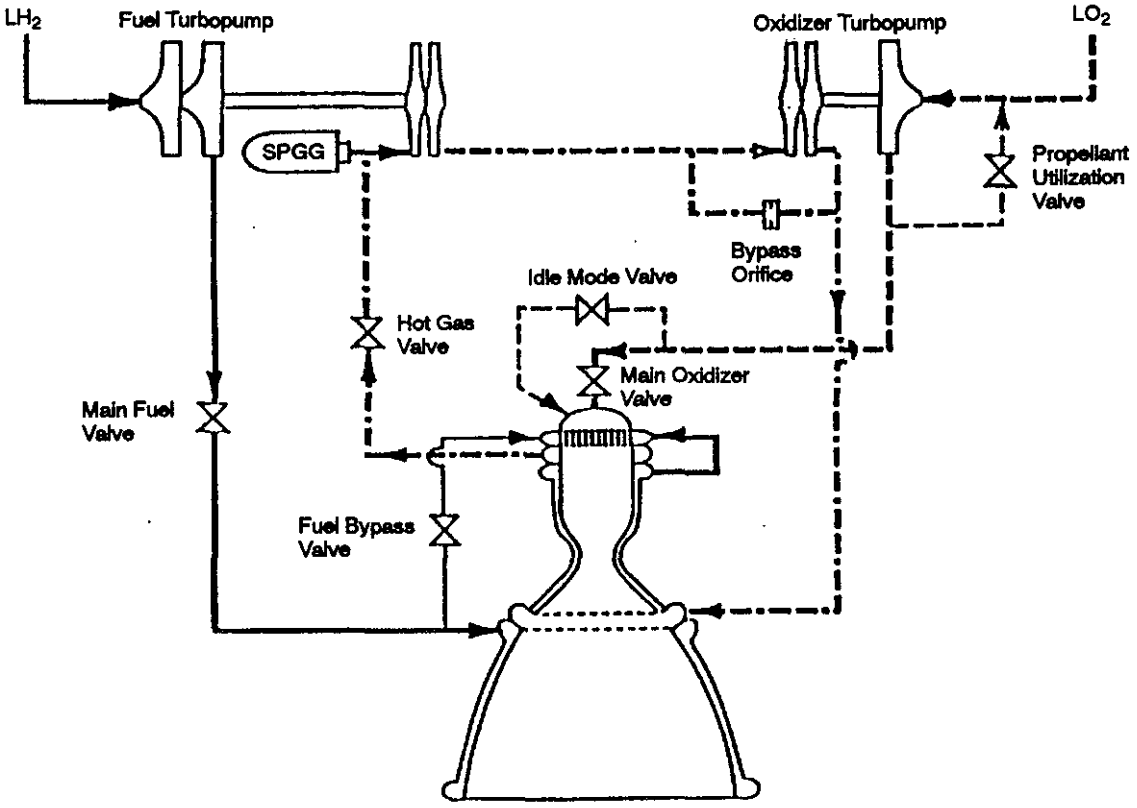


Fig. 2. The Innovative J-2S is the Largest Tap-off Cycle Engine Ever Built and Tested

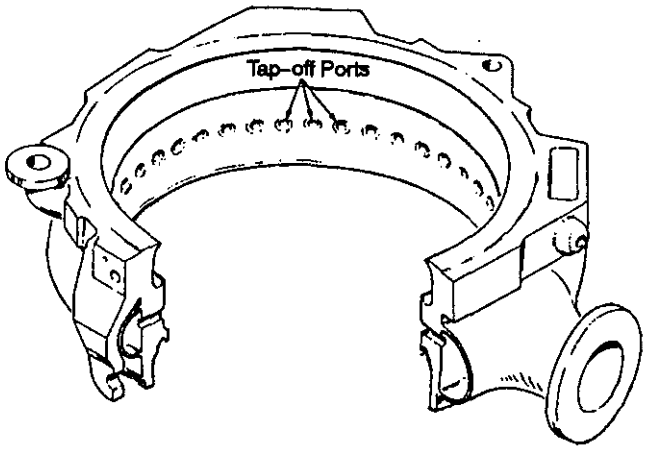


Fig. 3. J-2S Tap-off Ports Supplied Hot Turbine Drive Gas Without a Separate Gas Generator

Engine capability enhancements included the incorporation of throttling capability as well as low thrust idle mode operation. Throttling was accomplished by modulating the tap-off valve. This simple procedure was possible since gas generator flammability limit concerns were not applicable in this cycle. The production of the hot gas throttling valve was not a great challenge since it was not required to completely seal at closure. J-2S throttleability was demonstrated to as low as 6:1 at the Arnold Engineering Development Center (AEDC) J-4 test facility.

The idle mode of operation was originally intended to provide a propellant settling burn as well as provide rapid thermal conditioning of the engine. This pressure fed operating mode was kept stable by only supplying liquid oxidizer to the inner rows of the main injector by way of a separate manifold in the oxidizer dome. By reducing the effective oxidizer injector orifice area, stable combustion was attained at thrust levels as low as 6,000-lb thrust. In this operating mode, hydrogen was supplied to all of the injector elements to provide sufficient film cooling to the chamber such that no chamber erosion was observed. A summary comparison of the J-2 and J-2S is shown in Fig. 4.

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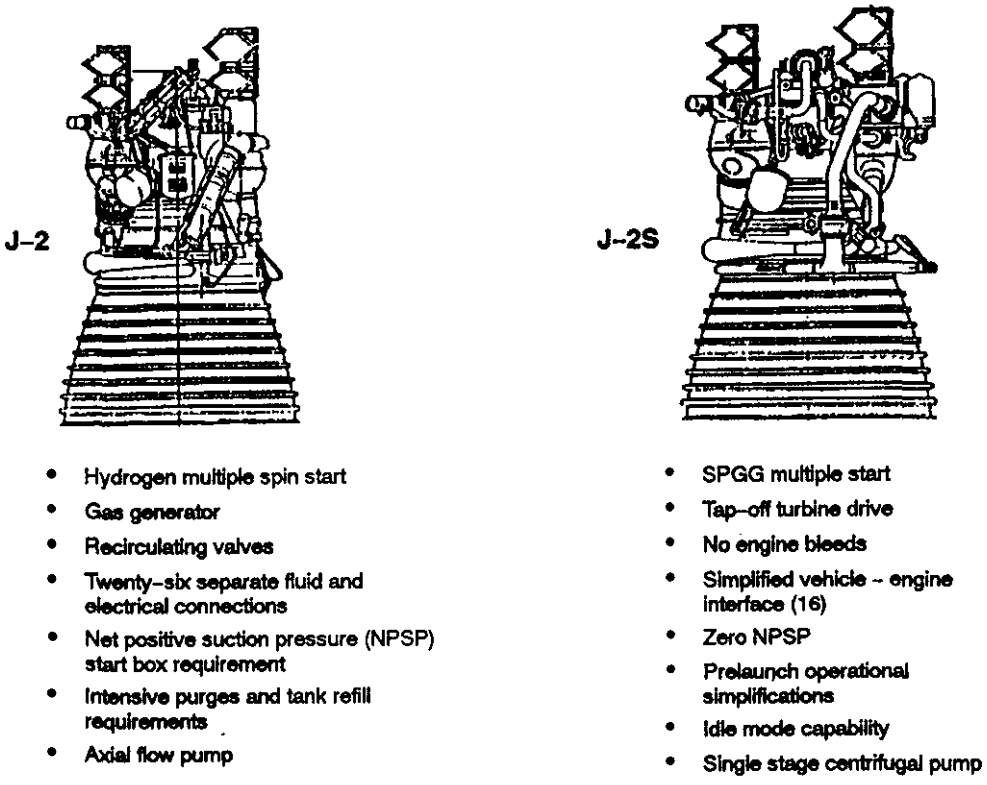


Fig. 4. The J-2S was Designed as a Simplified, Improved Performance Replacement for the J-2 Engine

Engine Development Program

Initial concept evaluation for the tap-off cycle was performed on a modified J-2 engine designated J-2X. The J-2X test bed demonstrated the feasibility of diluting the hot gas controllably to provide turbine drive gas without the need for a separate gas generator. This evolutionary approach has been cited as one of the key factors in the success of the J-2S development program.

Following testing of the J-2X test bed, actual flight configuration J-2S engines were tested at Rocketdyne's Santa Susana Field Laboratories (SSFL). This test program developed operation in idle mode as well as identifying and solving a fluid dynamic coupling phenomenon commonly identified with its characteristic frequency, 4,400 Hz.

Additional testing on the engine was performed at the AEDC to characterize the engines altitude operating characteristics. It was there in the J-4 test cell that the 6:1 throttling tests were conducted.

The engine development program was completed with six engines tested 273 times for a total of 30,858 sec, not counting the 10,756 sec tested on the J-2X test bed. A summary of the production configuration testing is shown in Table 1. At this point the NASA and Rocketdyne were ready to begin the two engine flight certification program. This engine development is a sharp contrast to the 31 development engines required for the J-2 engine and serves as an example of the efficiency derived from the evolutionary approach of engine development. The difficult lessons learned during the J-2 gas generator cycle engine development program were applied to the development of the J-2S tap-off cycle engine and provided substantial cost reduction while providing a technically superior product even though significant technology advances were injected into the program.

Production Restart

As part of Contract No. NAS8-39210, NASA Marshall Space Flight Center (MSFC) sponsored Rocketdyne to evaluate the requirements to restart production of the J-2S engine. During this study J-2S production drawings were reviewed to determine what changes would be required on the engine to allow production in a 1990s environment. Components were reviewed for producibility, material obsolescence (including environmental issues), reliability/operability concerns, as well as updating manufacturing techniques to state-of-the-practice.

An interesting finding of the study was that the engine could be produced from existing drawings. No material changes were required and all processes used in the original manufacture were still possible (though not necessarily practical). Only the engine

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Table 1. J-2S Accumulated Engine Test Log (Production Configuration Only)

Engine	Test Stand	Tests	Tank Head Idle Mode (sec)			Powered Idle Mode (sec)	Main Stage (sec)
			Pre-M/S	Post-M/S	Total		
J111	Delta-2A	29	1,503.1	30.0	1,533.1	0.0	827.0
J111	AEDC (J-4)	5	592.8	0.0	592.8	0.0	34.0
J112	Delta-2A	24	160.0	0.0	160.0	0.0	4,298.4
J112	AEDC (J-4)	57	3,649.1	59.3	3,708.4	252.4	352.6
J113	Delta-2A	99	190.0	40.0	230.0	0.0	11,508.8
J114	Delta-2A	19	0.0	0.0	0.0	0.0	4,445.4
J115	Delta-2A	3	0.0	0.0	0.0	0.0	402.5
J115	AEDC (J-4)	14	478.1	11.1	489.2	0.0	332.2
J116	Delta-2A	23	0.0	0.0	0.0	0.0	1,691.9
	Totals	273	6,573.1	140.4	6,713.5	252.4	23,892.8

electronics absolutely needed updating, and only because comparable components are no longer produced.

The study identified several changes that would greatly improve the producibility of the engine, such as adopting the use of castings in manifolds (including the tap-off manifold), the use of die forgings, and the incorporation of numerically controlled machining. It was found that several of the less complex components would benefit from complete redesign to fully benefit from these improved manufacturing processes.

The conclusion of the study was that the J-2S could be brought to a production-ready status in a relatively timely and economic manner. This conclusion is based on Rocketdyne's previous experience with restarting production of engines and includes a low risk six engine demonstration-certification test program. The first production engines would be available for delivery 4 years after authority to proceed.

Engine Uprating Potential

As the development of the J-2S was under way, studies were performed to assess the feasibility of increasing the thrust and specific impulse of the engine. This type of uprating was under continual study in other existing engine programs since greater vehicle capability was always a goal.

The studies on the J-2S examined the engine's uprate potential using the classic method of increasing turbomachinery output to increase chamber pressure. This did not necessitate extensive redesign of components, merely increased structure at certain critical locations determined analytically. This type of uprate procedure had been used successfully on the J-2, F-1, and MA-3, along with several engines that would follow.

The results of these studies on the J-2S was that the engine turbomachinery and thrust chamber could withstand uprating to over 320,000 lb of vacuum thrust with only minor modifications to retain structural margins. Modifications in the turbomachinery consisted of strengthening turbine blades and manifolds, the oxidizer volute and impeller, and the fuel pump volute. Also required would be the strengthening of the turbine exhaust gas manifolds and main valve housings, as well as changing the thrust chamber forward manifolds to Inconel 718 and enlarging the main valve actuators. No high risk component reconfiguration effort is necessary. This result is consistent with the experience base with open cycle engines. Greater tap-off flows are readily available to provide the increased turbopump power requirements without perturbing the basic operation of the remainder of the engine cycle.

Improvement of engine specific impulse is also possible since the original J-2S was constrained to the envelope restrictions of the basic J-2 engine. If a 105:1 nozzle were used, rather than the baseline 40:1 nozzle, the J-2S would match the specific impulse performance of the Space Shuttle Main Engine (SSME). To achieve this operating point however, the J-2S would be 40 in. longer and 34 in. wider since it operates at a considerably lower chamber pressure. This simply illustrates the range of feasible operating points possible with this engine.

Table 2 shows a comparison of the J-2 with the J-2S in its baseline configurations as well as with some of the performance option configurations.

Conclusions

The J-2S represents one of the more unusual case studies in the history of rocket engines. It is often thought of as a simple evolution from the J-2 engine, yet it operates with a different

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Table 2. J-2S Provides a Range of Performance Options to Meet Various Vehicle Requirements

Engine	J-2	J-2S					
	Baseline	Baseline	Performance Options				
			Nozzle Replacement		Uprating		
Expansion Area Ratio	27.5:1	40:1	80:1	105:1	40:1	80:1	105:1
Vacuum Thrust (lb)	230K	265K	272.5K	275K	320K	327.5K	330K
Vacuum Specific Impulse (sec) (at 5.5:1 MR)	425.0	436.0	448.3	452.8	434.5	446.8	451.4
Weight w/Accessories (lb)	3,454	3,800	3,755	3,855	4,120	4,040	4,200
Length from Gimbal Plane (in.)	116	116	172	196	116	172	196
Exit Diameter (in.)	80	80	112	128	80	112	128

engine cycle, uses new design turbomachinery, and incorporates unique operating capabilities such as deep throttling and idle mode. The thrust and performance growth were within previously demonstrated cycle upgrade limits, however the J-2S achieved these without expanding the engine envelope. The engine was simplified while the operational characteristics were improved. All this was accomplished with a six engine development program before it was ready to enter flight certification.

This serves as an excellent example of the innovation that is possible when the lessons learned of a previous development program are applied rigorously to any follow-on effort. Some of the key elements that allowed this type of success were an early understanding of requirements (J-2 envelope and interfaces), a small dedicated customer/contractor team, and the availability of experienced personnel.

The J-2S engine was never certified or flown, a victim of the Saturn program cancellation. It has reappeared in numerous booster engine trade studies in applications ranging from the space shuttle to advanced expendable launch systems to single-stage-to-orbit vehicle concepts. Other studies show this engine as the baseline upper stage engine for some phases of proposed lunar-Mars missions. Whatever the application, the J-2S remains a viable engine option within this thrust class over 20 years after its development.

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